area, as part of the Windows Server (WS) 2012 R2 effort, we have significantly improved Linux support and are making it possible for Linux workloads to fully leverage the benefits of the Microsoft platform and the ecosystem. Some of the key areas of Linux investment in the WS 2012 R2 wave were: full dynamic memory support, multichannel support for performance critical devices, and host initiated backup for Linux guests.

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Q Can you start with how Microsoft and the Linux community are working together on kernel contributions?

It's all part of our larger vision, which is to ensure that anything our customers want to run, they can run on Windows Server Hyper-V and in Windows Azure. This is what customers tell us they want.

One of the best things about getting to this point has been going through the process of working with the Linux community to refine and improve our code so that it best serves our customers. It was particularly gratifying to hear from Linux Foundation VP Greg Kroah-Hartman (in an interview with Wired magazine) that merging Microsoft's code into the Linux kernel tree made it smaller and easier to maintain.

Q Being a contributor to the Linux kernel, how do you react to Linus' worries on how Linux will handle the end of Moore's Law?

The laws of physics dictate that Moore's Law will have to end and the question is not 'if' but rather 'when'. Future innovation in computing is going to be in building and managing massively distributed computing infrastructures. Advances in performance and scalability are not going to be based on higher clock speeds but rather on new application paradigms, where we can bring to bear massive computational power based on commodity hardware. As network bandwidth improves, we can deliver a rich user experience on a variety of client devices served from a cloud of compute and storage infrastructure. Linux is well positioned in this space.

As we get to the physical limits on a single device, we may have to restate Moore's Law to take into account the new paradigm for increasing computational capacity.

Q In 2012, Microsoft was one amongst the top 20 companies contributing to the Linux kernel. But this year it is not in the Top 20 list. What are your views on this change?

This has been a topic of some discussion since the Linux conference in New Orleans back in September. There was a spike in activity when the Hyper-V drivers were initially contributed to the Linux kernel and went through the acceptance process with the Linux community. Since then, Microsoft has contributed at a steady rate with ongoing improvements to Linux support on Hyper-V. We will contribute on an ongoing basis as Hyper-V introduces new capabilities and we bring those capabilities to Linux guests.

We at Microsoft have always been focused on addressing customer needs. Our investments in supporting Linux are to ensure that Linux is, and continues to be, a first class environment on our platforms. Indeed, I would submit to you that the impact of our contributions in the last year was every bit as significant as what we did in the previous year from a Linux customer's perspective.

Q In the coming months, what will you be focused on with respect to kernel development?

We are in the process of ensuring that all of our WS 2012 R2 Linux related investments can be delivered through various commercial Linux distributions. Additionally, I am working on a couple of cool features that I am not sure I can disclose here.

My team remains committed to ensuring as far as possible that whatever features we implement in Hyper-V and Windows Azure are available to Linux guests. I want to get to the point where customers who are considering deploying Linux will see the Hyper-V and Windows Azure environments as the best choice for running Linux.

For example, we've just finished adding dynamic memory for Linux guests, which is a very important feature when a customer wants to consolidate a lot of workloads. It will allow you to manage the physical memory better across virtual machines. There are also a range of other drivers and interoperability between Linux and Windows that we'll be working on; basically, we want to look at anything that we feel our customers do or will find important.

Q What would your suggestions be to a new developer who wants to contribute to the Linux kernel?

Linux kernel development is truly open and anybody with good programming skills can contribute. If somebody is interested and is new to the kernel, I would suggest starting work on cleaning up drivers in the Linux kernel staging area: drivers/staging/. The drivers in the staging directory have numerous issues that range from being stylistic to